DEMO: Wireless Attacks on Aircraft Landing Systems

Harshad Sathaye, Domien Schepers, Aanjhan Ranganathan, and Guevara Noubir
Northeastern University, Boston, USA

ABSTRACT

Modern aircraft heavily rely on several wireless technologies for communications, control, and navigation. In this work, we demonstrate the vulnerability of aircraft instrument landing systems to wireless attacks. We show that it is possible to fully and in fine-grain control the course deviation indicator, as displayed by the ILS receiver, in real-time, and demonstrate it on aviation-grade ILS receivers. We develop a tightly-controlled closed-loop ILS spoofer that autonomously adjusts the adversary's transmitted signals based on the aircraft's GPS location to cause an undetected off-runway landing. We demonstrate the integrated attack on an FAA certified flight-simulator (X-Plane)'s AI-based auto-land feature and show success rate with offset touchdowns of 18 meters to over 50 meters.

CCS CONCEPTS

• **Security and privacy → Mobile and wireless security**.

ACM Reference Format:
Harshad Sathaye, Domien Schepers, Aanjhan Ranganathan, and Guevara Noubir. 2019. DEMO: Wireless Attacks on Aircraft Landing Systems. In *WiSec '19: ACM Conference on Security and Privacy in Wireless and Mobile Networks, May 15–17, 2019, Miami, FL, USA.* ACM, Miami, FL, USA, 3 pages. https://doi.org/10.1145/3317549.3326298

1 INTRODUCTION

Today, the aviation industry is experiencing significant growth in air traffic with more than 5000 flights [6] in the sky at any given time. Typically, air traffic control towers handle more than a thousand takeoffs and landings every day. For example, Atlanta's Hartsfield-Jackson International airport handles around 2500 takeoffs and landings every day. The modern aviation ecosystem heavily relies on a plethora of wireless technologies.

One of the most critical phases of an airplane's flight plan is the final approach or landing phase as the plane descends towards the ground actively maneuvered by the pilot. For example, 59% of the fatal accidents documented by Boeing [7][1] occurred during descent, approach and landing. Several technologies and systems such as GPS, VOR, DME assist the pilot in landing the aircraft safely. The Instrument Landing System (ILS) [1] is today the de-facto approach system used by planes at a majority of the airports as it is the most precise system capable of providing accurate horizontal and vertical guidance. At Boston's Logan International Airport, 405,822 [2] flight plans were filed in 2017. 95% flight-plans were instrument flight rule (IFR) plans. Instrument flight rules are a set of instructions established by the FAA to govern flight under conditions in which flying by visual reference is either unsafe or just not allowed. ILS incorporates radio technology to provide all-weather guidance to pilots which ensures safe travel and any interference can be catastrophic. With increasing reliance on auto-pilot systems and widespread availability of low-cost software-defined radio hardware, adversarial wireless interference to critical infrastructure systems such as the ILS cannot be ruled out. In this work, we demonstrate the security of aircraft instrument landing system against wireless attacks. We analyze the ILS waveforms, the transmitters and receivers, and show that ILS is vulnerable to signal spoofing attacks. We generate specially crafted radio signals similar to the legitimate ILS signals using low-cost software-defined radio hardware platform and successfully induce aviation-grade ILS receivers, in *realtime*, to lock and display arbitrary alignment to both horizontal and vertical approach path. This demonstrates the potential for an adversary to the least be able to trigger go-arounds, causing air traffic disruption, and in the worst case, cause the aircraft to overshoot the landing zone or miss the runway entirely. In order to evaluate the complete attack, we develop a tightly-controlled closed-loop ILS spoofer. It adjusts the adversary's transmitted signals as a function of the aircraft GPS location, maintaining power and deviation consistent with the adversary's target position, causing an undetected off-runway landing. We demonstrate the integrated attack on an FAA certified flight-simulator (X-Plane), incorporating a spoofing region detection mechanism, that triggers the controlled spoofing on entering the landing zone to reduce detectability. We systematically evaluate the performance of the attack against X-Plane's AI-based autoland feature, and demonstrate the systematic success rate with offset touchdowns of 18 meters to over 50 meters.

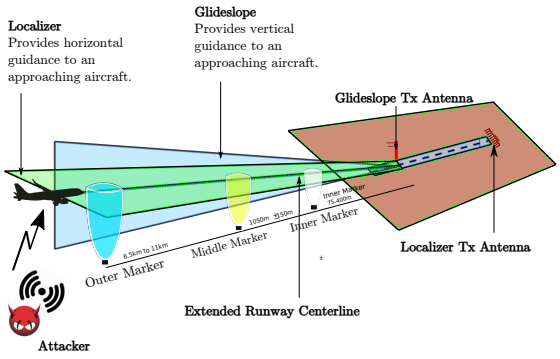


Figure 1: Overview of ILS sub-systems. The ILS consists of three subsystems: i) Localizer, ii) glideslope, and (iii) marker beacons.

[1]https://www.boeing.com/resources/boeingdotcom/company/about_bca/pdf/statsum.pdf

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WiSec '19, May 15–17, 2019, Miami, FL, USA
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ACM ISBN 978-1-4503-6726-4/19/05.
https://doi.org/10.1145/3317549.3326298

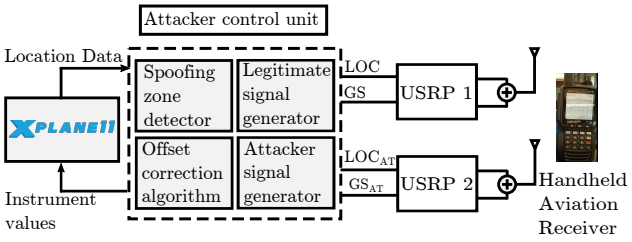


Figure 2: Schematic of the experiment setup used for evaluating the attacks on ILS.

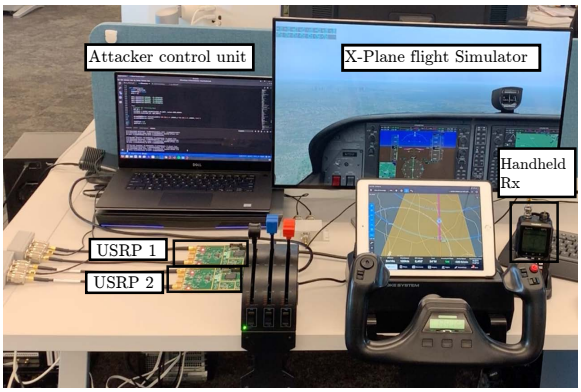


Figure 3: Photo of the experiment setup.

2 INSTRUMENT LANDING SYSTEM

ITU defines ILS [10] as "a radio navigation system which provides aircraft with horizontal and vertical guidance just before and during landing and at certain fixed points, indicates the distance to the reference point of landing". Autopilot systems on some modern aircraft [9] use ILS signals to execute a fully autonomous approach and landing, especially in low visibility settings. ILS (Figure 1) comprises of three independent subsystems: i) localizer, (ii) glideslope and iii) marker beacons. The localizer and the glideslope guide the aircraft in the horizontal and vertical plane respectively. The marker beacons act as checkpoints that enable the pilot to determine the aircraft's distance to the runway. With the advent of GPS and other localization technologies, the marker beacons are less important today and increasingly obsolete. However, the localizer and the glideslope play a major role in an aircraft's safe landing today and is expected to remain so for many years.

2.1 ILS Signal Generation

ILS signal generators leverage *space modulation* i.e., use multiple antennas to transmit an amplitude modulated radio frequency signal with various powers and phases. The transmitted signals combine in the airspace to form signals with different depths of modulation (DDM) at various points within the 3D airspace. Each DDM value directly indicates a specific deviation of the aircraft from the correct touchdown position. For example, the signals combine in space to produce a signal with zero DDM along the center-line of the runway. It is important to note that unlike traditional modulation techniques where the modulation occurs within the transmitter, in space modulation, the signals mix within the airspace.

The process of generating the localizer and glideslope signals are similar with differences in the carrier frequency, depth of modulation of 90 Hz and 150 Hz tones and how they are combined in space. The depth of modulation or modulation index is the measure of the extent of amplitude variation about an un-modulated carrier. The depth of modulation is set at 20% and 40% respectively for localizer and glideslope signals. The output of both the 90 Hz and 150 Hz modulator is then combined to yield two radio frequency signals: a carrier-plus-sidebands (CSB) which contains both the sideband energy and the carrier energy and a sidebands-only (SBO) signal where carrier component is suppressed. The CSB and SBO signals are transmitted with carefully chosen phase shifts such that when they combine in space, the resulting signal enables the aircraft to determine its alignment with respect to the centerline and the glidepath angle.

2.2 ILS Receiver

The combined localizer and glideslope signals received at the aircraft are amplified, demodulated, and filtered to recover the 90 Hz and 150 Hz components. A bridge rectifier is used to convert the amplitude of the recovered tones to DC voltage levels. The DC voltage output is directly proportional to the depth of the modulation of the 90 Hz and 150 Hz tones–a direct measure of the dominating frequency signal. The DC voltage causes the course deviation indicator needle to deflect based on the DDM of the two tones thereby precisely indicating the aircraft's lateral and vertical deviation from approach path. Note that localizer-glideslope frequencies occur in pairs and are hardcoded i.e. has to select just the appropriate localizer frequency and the receiver tunes to the correct glideslope frequency.

3 ILS SIGNAL SPOOFER

The ILS signal spoofer transmits specially crafted ILS signals at a power level such that the legitimate signals get overpowered by the attacker's signal at the receiver. The main reason why such an attack works is that the receivers "lock" and process only the strongest received signal. We note that the attacker signal can be specially crafted to force the CDI needle to indicate a specific offset. Recall that the ILS receiver on-board receives a mix of the transmitted CSB and SBO signals that contain the 90 and 150 Hz tones. The amplitude of received 90 and 150 Hz tones depends on the position of the aircraft relative to the runway and its approach path angle. For example, the 90 Hz tone will dominate if the aircraft is offset to the left of the runway and the 150 Hz dominates to the right. Similarly, for glideslope, the 90 Hz tone dominates glide angles steeper than the recommended angle, and the 150 Hz tone dominates otherwise. Both 90 and 150 Hz will have equal amplitudes for a perfectly aligned approach. Therefore, to execute an overshadow attack, it is sufficient to generate signals similar to the received legitimate ILS signals and transmit at a much higher power as compared to legitimate ILS signals. In other words, the attacker need not generate CSB and SBO signals separately; instead can directly transmit the combined signal with appropriate amplitude differences between

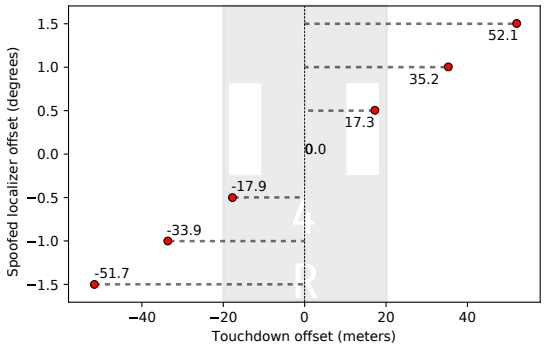


Figure 4: Results of localizer spoofing.

the 90 and 150 Hz tones. The amplitude differences are calculated based on the offset the attacker intends to introduce at the aircraft.

4 EXPERIMENTAL SETUP

Our experimental setup is shown in Figure 2 and Figure 3. The setup consists of four main components: i) X-Plane 11 flight simulator, ii) attacker control unit, iii) software-defined radio hardware platforms (USRP B210s) and iv) commercial aviation grade handheld navigation receiver. We use X-Plane 11 flight simulator to test the effects of spoofing attack on the ILS.

X-Plane is a professional flight-simulator which qualifies for FAA-certified flight training and is used by pilot training schools. It is capable of simulating various aircraft with accurate aerodynamic models and is also capable of simulating various weather conditions. X-Plane allows interaction with the simulator and instruments through a variety of mobile apps and UDP/IP networks. This feature allowed us to manipulate the instrument readings for evaluating our ILS attacks. Additionally, X-Plane has autopilot and AI-based autoland features which we leverage in our experiments. In other words, X-Plane contains all the features and flexibility to evaluate our proposed attacks in a close to the real-world setting. The second component of our setup is the attacker control unit module which takes the location of the aircraft as input from X-Plane and generates signals for the attack. The module is also responsible for manipulating X-Plane's instrument panel based on the effect of the spoofing signal on the receiver. The attacker control unit module is a laptop running Ubuntu and contains four submodules: spoofing zone detector, offset correction algorithm, legitimate signal generator, and attacker signal generator. The spoofing zone detector identifies whether an aircraft is entering its first waypoint of the final approach and triggers the start of spoofing. The spoofing zone detector is responsible for timely starting of the spoofing attack so as to prevent any abrupt changes in the instrument panel and therefore avoid suspicion. The offset correction algorithm uses the current location of the aircraft to continuously correct its spoofing signals taking into consideration aircraft's corrective actions. Note that the location data received from X-Plane can be analogous to receiving the location data through ADS-B signals [8] in the real world. The output of the offset correction algorithm is used to generate fake ILS signals. We also generate legitimate signals to evaluate the effect of overshadow and single-tone attacks. We use

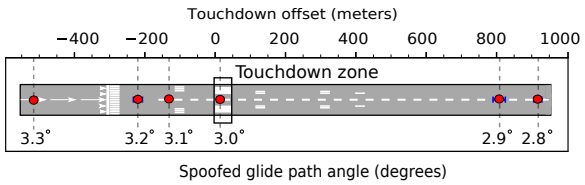


Figure 5: Results of glideslope spoofing.

two USRP B210s [3], one each for transmitting legitimate ILS signals and attacker signals. We conducted the experiments in both wired and wireless settings. We note that the wireless experiments were performed in controlled settings to prevent any signal leakage as it is illegal to transmit ILS signals over the air. We use two different ILS receivers, a Yaesu FTA-750L [5] and a Sporty's SP-400 Handheld NAV/COM Aviation [4] to evaluate the attacks. The spoofing signals were generated to fake various lateral and vertical offsets. Note that the fake signals were generated in real-time. The spoofing glideslope angles were between 2.8° and 3.3°. For each spoofing angle and offset, we performed five automated landings and the results are shown in Figure 4 and Figure 5. The aircraft landed with an 18 m offset from the runway centerline for a spoofing offset of just 0.5°. Note that this is already close to the edge of the runway and potentially go undetected by both the air-traffic controllers as well as pilots onboard, especially in low visibility conditions. In the case of glideslope, a shift in the glide path angle by 0.1° i.e., 2.9° glide path angle instead of the recommended 3°, caused the aircraft to land almost 800m beyond the safe touchdown zone of the runway. We have uploaded a video demonstration of the attack for reference (https://youtu.be/Wp4CpyxYJq4).

5 DEMONSTRATION

The demonstration includes a live demo of the attack on the flight simulator (as shown in Figure 3) which includes a fully functional attacker control unit with an offset correction algorithm for real-time fake signal generation and spoofing zone detector and necessary flight controls for X-Plane flight simulator. The demonstration will be accompanied by a poster.

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